

Threshold

1. A classifier (like logistic regression) often gives a probability score (e.g., 0.82).
2. The threshold decides how high that probability must be to classify the result as positive (1) instead of negative (0).
3. Default threshold = 0.5, but this is not always optimal!

How to Choose a Threshold in Binary Classification?

You don't want to blindly use threshold = 0.5. Instead, you want to:

- Choose the threshold on your training data, but not directly.
- Use a **validation method** like **cross-validation** or a **separate validation set** to avoid overfitting.

✅ Step-by-Step Strategy

◆ Step 1: Train-Test Split

Split your data into:

- **Training set** → used to train the model.
- **Validation set** (or use **cross-validation**) → used to **tune the threshold**.
- **Test set** (optional) → used for final performance check.

If using k-fold cross-validation, each fold plays the role of a validation set once.

◆ Step 2: Train the Model

Train your model on the training data. The model outputs **probabilities**.

◆ Step 3: Predict on Validation Data

Get predicted **probabilities** (not class labels) on the validation set.

◆ Step 4: Try Multiple Thresholds

Loop over many thresholds (e.g., 0.1 to 0.9) and:

- Convert probabilities to labels using each threshold.
 - Calculate **F1 score** for each.
 - Choose the **threshold that gives the highest F1 score**.
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◆ Step 5: Apply Threshold on Test Set

Now that you have the **best threshold** from the **validation set**, apply it to classify the **test set** (unseen data).

! Important Notes:

-  **Yes**, calculate F1 score on the **validation set or cross-validation folds**, NOT on the training set.
 -  Don't choose the threshold based on training set performance — this leads to **overfitting**.
 -  Once you've selected the threshold, you can **retrain on full training data** (train + val) and test on test data.
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Summary:

Task	Where Done
Train model	Training set
Choose threshold (e.g. best F1)	Validation set or CV
Evaluate final performance	Test set (optional)

F1 Score

✅ Why is the F1 Score Important?

1. Balances Precision and Recall

- **Precision** tells you how many predicted positives were actually correct.
- **Recall** tells you how many actual positives your model found.

If one is high and the other is low, **accuracy can be misleading**, but F1 will reflect the weakness.

2. Useful for Imbalanced Data

When classes are **uneven** (like 90% negative, 10% positive), **accuracy** can be very high even with a bad model. But **F1 score exposes this**.

Example: Predicting rare diseases, fraud detection, spam emails.

3. Helps with Model Comparison

F1 score gives you a **single value to compare models**, especially when you care equally about avoiding **false positives** and **false negatives**.

📌 When to Use F1 Score

- Your data is **imbalanced**.
 - **False positives and false negatives** are both costly.
 - You want a **single metric** that reflects both **precision and recall**.
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❌ When NOT to Use F1 Score

- If **only precision** or **only recall** matters.
- If you have **balanced data** and want a simpler metric like **accuracy**.

How Precision and Recall Change with Threshold

Threshold	Precision	Recall	Use Case
Low	Low	High	Medical, fraud detection
High	High	Low	Spam filters, security alarms
Optimal	Balanced	Balanced	Use F1 score to find this

Lower threshold (e.g., 0.3):

- ❖ More items are classified as **positive**.
- ❖ ▲ **Recall** increases (you catch more real positives).
- ❖ ▼ **Precision** decreases (more false positives).

Higher threshold (e.g., 0.7):

- ❖ Fewer items are classified as **positive**.
- ❖ ▲ **Precision** increases (less false positives).
- ❖ ▼ **Recall** decreases (misses some actual positives).

Using Precision-Recall Curve

You can:

1. **Plot precision vs recall** for different thresholds.
2. Choose the threshold that gives you the **best tradeoff** (depends on your goal).

✓ For example:

- In spam detection: you might **prefer high precision** (don't mark good emails as spam).
- In disease detection: you might **prefer high recall** (catch all possible cases, even with some false alarms).

Using F1 Score to Choose Threshold

You can also:

1. Calculate **F1 score** at different thresholds.
2. Choose the **threshold where F1 is highest** (i.e., where precision and recall are best balanced).